

Ganxuan Yang

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HIGHER EDUCATION

Lancaster University & Beijing Jiaotong University

2023.9-2027.6

Bsc Computer Science (Joint Programme)

RESEARCH INTERESTS

I am boardly interested in Security, Cryptography, AI-based Aerial Robotics, Embodied Intelligence, AI for healthcare and Generative models for interactive visual media, especially game and animation.

RESEARCH EXPERIENCES

BRASP: Boolean Range Queries over Encrypted Spatial Data with Access and Search Pattern Privacy
(<https://arxiv.org/abs/2604.07797>)

Supervisor: Muhammad Bilal (Lancaster University) **Role:** Co-first author

Individual Contributions:

- Co-designed BRASP's dual-server architecture for access and search-pattern privacy, integrating TPF, TUR, and Hilbert-curve spatial indexing.
- Implemented the cryptographic modules, encrypted index pipeline, core protocols, and distributed prototype across the Data Owner, cloud servers, and client.
- Designed and conducted performance experiments under varying dataset sizes, query workloads, and updates, and released the complete system as open-source software.

PROJECT

Unity-based individual game project "Eclipse Strike"

2026.3-2026.6

- Fully-developed Photon Engine-Driven and FPSMUL-based onlile multiplayer FPS set, featuring asymmetric factions, objective-driven combat, vehicles, destructible defences, low-gravity environments, and real-time voice communication.
- Developed reinforcement-learning-driven autonomous NPC agents capable of learning behaviours, strategies, and adaptive decision-making through interactions with the game environment.

Internship

Hunan Shuxin Intelligence Technology Co., Ltd.

2026.7-Now

Central South University and Xiangya Hospital frontier project

- Contributed to an AI-enabled wearable health-monitoring research platform informed by interdisciplinary research at Central South University and clinical expertise from Xiangya Hospital.
- Implemented full-stack and mobile modules for sensor-data acquisition, processing, visualization, and cloud synchronization, supporting continuous monitoring and downstream data analysis.
- Supported embedded-device integration and prototype validation through communication debugging and end-to-end testing across wearable devices, backend pipelines, and user applications.

Tianjin Hanyu Haode Technology Development Co., Ltd.

2025.7-2025.8

Sino-Japanese joint venture with Denpa Jikkensha Co., Ltd. (now Denpa-sha)

- Contributed to the development of UAV flight-control and six-degree-of-freedom hydraulic motion-simulation systems, implementing real-time control logic, telemetry processing, and software-hardware interfaces.
- Conducted integration testing and prototype validation by analysing discrepancies between simulated flight states and physical platform responses, supporting embedded-device debugging and system refinement.

Skills

Languages: Proficient in Mandarin and English, Basic Japanese

Programming Skills: C, C#, Java, Python, SQL, Go, Flutter, React, Matlab, Haskell, LaTeX